

Notes: Marra (JPE, 2025)

Estimating an Auction Platform Game with Two-Sided Entry

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1 Big picture

- **Question.** How should an auction platform allocate fees across buyers and sellers when entry on both sides of the market is endogenous and users create network effects for each other?
- **Setting.** The paper studies an online UK wine auction platform, Bid for Wine, where individual sellers list heterogeneous bottles and potential bidders decide whether to inspect and enter listings.
- **Core challenge.** In a two-sided auction platform, entry by sellers changes not only the number of listings but also the composition of listings. If sellers are heterogeneous and know their values before entry, fee changes alter who comes to market. That, in turn, changes reserve prices, bidder entry, transaction prices, and welfare incidence.
- **Main theoretical contribution.** The paper develops a static auction platform game with **two-sided entry** and **selective seller entry**. A key feature is a “lemons effect”: excluding high-value sellers can raise bidder surplus because those sellers set higher reserve prices.
- **Main econometric contribution.** The paper shows that the model primitives are nonparametrically identified with basic auction data and proposes a computationally feasible estimator that resembles a nested pseudo-likelihood algorithm.
- **Main empirical results.** A £1 increase in the listing fee lowers the expected surplus of sellers who remain on the platform by only about £0.54–£0.78, not by the full £1, because high-value sellers exit and bidder entry rises on the remaining listings. Pairing the fee increase with a bidder subsidy can make most users, including many sellers, better off despite higher seller fees. Alternative fee structures can raise platform revenue by up to about 40%.
- **Why this paper matters.** It shows that standard one-sided intuitions about fee incidence and antitrust damages can be badly misleading in two-sided platform markets. In particular, welfare and harm must be evaluated jointly across both sides.

2 Data and measurement (key objects)

2.1 Observed auction-level panel

- **Platform.** Bid for Wine (BW), an online UK peer-to-peer wine auction platform, with more than 20,000 users.
- **Sample construction.** The author scraped all open auctions every 30 minutes from January 2017 to May 2018.
- **Unit of observation.** A listing-auction.
- **Raw sample.** 3,481 auctions after excluding consignment auctions, spirits, and multiple-lot sales.
- **Key observed outcomes.** Hammer price, reserve price, sale indicator, number of bidders, number of bottles, shipping and payment conditions, seller ratings, storage conditions, in-bond status, ullage/fill level, and detailed text descriptions.

- **Main sample.** About 80% of auctions fall in the lowest seller-commission bracket (hammer price and reserve price at most £200). These are the paper’s baseline empirical focus.
- **High-end sample.** Auctions with hammer prices between £200 and £800 and reserve prices at most £800 are estimated separately to study heterogeneity across product classes.

2.2 Key measured objects

- **Average auction statistics.** Mean hammer price is about £140, mean number of bidders is 3.10, about 64% of listings sell, and about 67% have some reserve price.
- **Reserve-price measurement.** Public reserve prices are observed directly through raised minimum bids. Secret reserves are approximated from the highest standing price at which “reserve not met” is displayed and the lowest price at which it is met.
- **Seller population.** Of 15,762 registered users, 2,581 are potential sellers in the sense of having listed at least once, but only 263 list during the sample period.
- **Bidder population.** 10,856 bidder identities are observed indirectly through bidding activity, though bidder identities are generally not known. Among 246 feedback-leaving winning bidders, 41 also listed wine for sale.
- **Thin-market evidence.** Even with coarse product definitions, half of product-month cells have only one very similar listing, and another 30 percentage points have only two or three. This matters because the platform is not a thick commodity-like market.

2.3 Fee structure

The platform fee vector is

$$c = \{c_B, c_S, c_L\},$$

where c_B is the buyer premium, c_S is the seller commission, and c_L is the flat listing fee.

In the data:

- buyer premium: $c_B = 0$,
- seller commission: $c_S = 0.102$ for sale prices up to £200 and 0.09 for £200.01–£1,500,
- listing fee: $c_L = £2.10$.

The paper abstracts from the separate reserve-price fee and focuses on $\{c_B, c_S, c_L\}$.

2.4 Unobservables and timing

- **Bidder values.** Potential bidders draw idiosyncratic values for the wine but, in the baseline model, learn them only after entering a listing.
- **Seller values.** Sellers know their own idiosyncratic value (or marginal cost of selling) before deciding whether to list.
- **Timing.** The game has two stages:

- stage 1: sellers decide whether to list and bidders decide whether to enter listings,
- stage 2: sellers choose reserve prices and bidders submit bids.
- **Observed heterogeneity.** Auction observables Z are summarized through a quality term $q = e^{g(Z)}$, which scales values multiplicatively.

3 Models and empirical specifications (all equations + interpretation)

3.1 Two-sided IPV environment

Potential bidders draw values $V \sim F_V$ and potential sellers draw values $V_0 \sim F_{V_0}$. The model assumes two-sided independent private values:

$$V_i \perp V_{i'} \quad (i \neq i'), \quad V_i \perp V_{0k}.$$

Observed item characteristics enter through a common quality term:

$$\tilde{V}_i = qV_i, \quad \tilde{V}_{0k} = qV_{0k}, \quad q \perp (V_i, V_{0k}).$$

- **Interpretation.** This lets the empirical work partial out observed heterogeneity while preserving tractable auction-stage formulas in “homogenized” value space.

3.2 Large-population approximation and bidder counts

The number of bidders per listing is assumed Poisson:

$$f_{N_r}(n; \lambda_r) = \exp(-\lambda_r) \frac{\lambda_r^n}{n!}, \quad r \in \{r = 0, r > 0\}.$$

Here $r = 0$ denotes no-reserve auctions and $r > 0$ denotes positive-reserve auctions.

- **Interpretation.** The equilibrium on the bidder side is summarized by the mean number of bidders per listing, $\lambda_{r=0}^*$ and $\lambda_{r>0}^*$.
- **Why it matters.** This assumption makes the two-sided entry game computationally feasible and fits the observed bidder-count distribution well.

3.3 Auction-stage bidding and reserve pricing

With buyer commission c_B , a bidder with value v bids

$$b^*(v) = \frac{v}{1 + c_B}.$$

A seller with value $v_0 \geq v_0^R$ chooses an optimal reserve price $r^* = r^*(v_0)$ satisfying

$$r^* = \frac{v_0}{1 - c_S} + \frac{1 - F_V((1 + c_B)r^*)}{(1 + c_B)f_V((1 + c_B)r^*)}.$$

Sellers with $v_0 < v_0^R$ choose a zero reserve.

- **Interpretation.** The buyer commission scales down bids. The seller commission raises the optimal reserve. High-value sellers optimally set higher reserves.
- **Economic force.** Seller heterogeneity matters because reserve prices become an endogenous selection device affecting bidder entry.

3.4 Listing-level expected surplus without reserve prices

For zero-reserve auctions, the expected surplus per entering bidder when there are $n \geq 1$ bidders is

$$pi_B^{r=0}(n) = \frac{1}{n} \mathbb{E}[V_{n:n} - V_{n-1:n}].$$

For sellers, expected surplus is

$$pi_S^{r=0}(n, v_0) = (1 - c_S) \mathbb{E} \left[\frac{V_{n-1:n}}{1 + c_B} \right] - v_0.$$

- **Interpretation.** In no-reserve auctions, bidder surplus does not depend on seller type. This is why the two-sidedness of the market shows up much more sharply in positive-reserve auctions.

3.5 Listing-level expected surplus with positive reserve prices

For positive-reserve auctions, bidder surplus when there are $n \geq 1$ bidders and seller type v_0 is

$$p_B^{r>0}(n, v_0) = \frac{1}{n} \mathbb{E}[V_{n:n} - \max\{V_{n-1:n}, \tilde{r}\} \mid V_{n:n} \geq \tilde{r}] [1 - F_V(\tilde{r})],$$

where $\tilde{r} = (1 + c_B)r^*(v_0)$ is the buyer-premium-adjusted reserve.

Seller surplus is

$$p_S^{r>0}(n, v_0) = \left[(1 - c_S) \mathbb{E}[\max\{V_{n-1:n}, \tilde{r}\} \mid V_{n:n} \geq \tilde{r}] - v_0 \right] [1 - F_V(\tilde{r})].$$

- **Interpretation.** Bidder surplus is decreasing in both n and v_0 . The latter is the key selection channel: higher-value sellers set higher reserves, which make listings less attractive to bidders.

3.6 Entry equilibrium: zero reserve auctions

Expected bidder surplus from entering a zero-reserve listing when mean bidder count is $\lambda_{r=0}$ is

$$\Pi_B^{r=0}(\lambda_{r=0}) = \sum_{n \geq 1} p_B^{r=0}(n) f_{N_{r=0}}(n; \lambda_{r=0}) - e_{B,r=0}.$$

The equilibrium mean number of bidders per zero-reserve listing solves

$$\Pi_B^{r=0}(\lambda_{r=0}^*) = 0.$$

- **Interpretation.** This is standard free-entry logic on the bidder side.

3.7 Entry equilibrium: positive reserve auctions and seller selection

Given a candidate seller entry threshold $\bar{v}_0$, bidder surplus from entering a positive-reserve listing is

$$\Pi_B^{r>0}(\bar{v}_0; \lambda_{r>0}) = \sum_{n \geq 1} \left[\int_{v_0^R}^{\bar{v}_0} p_B^{r>0}(n, v_0) \frac{f_{V_0}(v_0)}{F_{V_0}(\bar{v}_0) - F_{V_0}(v_0^R)} dv_0 \right] f_{N_{r>0}}(n; \lambda_{r>0}) - e_{B,r>0}.$$

For any candidate threshold, the bidder side implies

$$\Pi_B^{r>0}(\bar{v}_0; \lambda_{r>0}) = 0,$$

which defines $\lambda_{r>0}^*(\bar{v}_0)$.

On the seller side, expected surplus for a seller of type v_0 is

$$\Pi_S^{r>0}(v_0; \lambda_{r>0}^*(\bar{v}_0)) = \sum_{n \geq 1} p_S^{r>0}(n, v_0) f_{N_{r>0}}(n; \lambda_{r>0}^*(\bar{v}_0)) - c_L - e_S.$$

The equilibrium seller entry threshold v_0^* solves

$$\Pi_S^{r>0}(v_0^*; \lambda_{r>0}^*(v_0^*)) = 0.$$

- **Interpretation.** Sellers use a threshold strategy: all types below v_0^* enter.
- **Key uniqueness result.** A higher seller-entry threshold worsens the reserve-price distribution seen by bidders, lowers bidder entry, and therefore lowers seller profits. This single-crossing property yields a unique equilibrium despite two-sided feedback.
- **Lemons effect.** The negative externality from higher-value sellers arises because they push up latent reserve prices and discourage bidders.

3.8 Nonparametric identification from order statistics and reserve prices

Bidder values are identified from zero-reserve auctions. The distribution of the second-highest order statistic satisfies

$$F_{V_{n-1:n}}(v) = n(n-1) \int_{\underline{v}}^v F_V(u)^{n-2} [1 - F_V(u)] du,$$

so inverting this relationship separately by n identifies F_V .

Seller values on the relevant support are identified by inverting the reserve-price rule:

$$v_0(r) = (1 - c_S)r - \frac{1 - F_V(r(1 + c_B))}{(1 + c_B)f_V(r(1 + c_B))}.$$

Hence the distribution of implied seller values among entering positive-reserve sellers is

$$F_{v_0(r)}(v_0) = \frac{F_{V_0 \geq v_0^R}(v_0)}{F_{V_0 \geq v_0^R}(v_0^*)}, \quad v_0 \in [v_0^R, v_0^*].$$

Entry costs are then recovered from the zero-profit conditions:

$$\Pi_S^{r>0}(v_0^*; \lambda_{r>0}^*(v_0^*)) = 0, \quad \Pi_B^{r=0}(\lambda_{r=0}^*) = 0, \quad \Pi_B^{r>0}(v_0^*; \lambda_{r>0}^*) = 0.$$

- **Interpretation.** Identification uses only standard auction observables plus reserve prices and bidder counts.
- **Main message.** The model primitives are identified without separate exclusion restrictions of the type often used in reduced-form two-sided market work.

3.9 Estimation method and parameterization

- The paper first estimates an auction-quality shifter $q_t = e^{g(Z_t)}$ from observables such as wine type, region, vintage, storage, shipping, seller ratings, and text-description content.
- Bidder and seller value distributions are parameterized as log generalized Gaussian distributions with location, scale, and shape parameters.
- Seller-value estimation is nested with the equilibrium seller-entry threshold because the support of observed reserve prices depends on v_0^* .
- The resulting algorithm resembles the Aguirregabiria-Mira nested pseudo-likelihood approach: iterate between structural parameters and the equilibrium threshold until convergence.
- An extra parameter $p_{0,r>0}$ is introduced so the model can fit the relatively large mass of positive-reserve listings with zero actual bidders.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in a naive interpretation

- A one-sided auction model that treats seller entry as fixed misses a first-order margin of adjustment: changing fees changes which sellers show up.
- Reduced-form incidence formulas that assume buyers are unaffected by seller-side fees fail because seller-side selection changes bidder entry and sale probabilities.
- Benchmark antitrust rules of thumb—for example, that winning bidders are unaffected by commission changes, or that harm should be allocated pro rata by direct overcharges—are not valid in this environment.

4.2 What identifies the model in the new approach

- **Bidder-side identification.** Zero-reserve auctions identify the bidder value distribution through second-highest bids and observed bidder counts.
- **Seller-side identification.** Positive-reserve auctions identify the seller value distribution on the relevant support through the reserve-price first-order condition.
- **Entry-cost identification.** Zero-profit conditions for the marginal seller and marginal bidders identify seller and bidder entry costs.
- **Equilibrium uniqueness.** The strict decrease of bidder entry in the seller threshold and the strict decrease of seller profits in own type give a unique fixed point, which makes the estimation problem tractable.

4.3 Why the empirical setting is especially useful

- The platform is a thin market with highly heterogeneous goods, so listing inspection is naturally costly.
- Sellers plausibly know their values ex ante because they already own the wine and often have owned it for years.
- Bidders plausibly learn values after inspection because each listing contains idiosyncratic information about provenance, fill level, storage, shipping, and seller credibility.
- Rich observables explain a large share of price variation, which reduces concerns that unobserved heterogeneity is the main driver of the results.

4.4 Relation to the literature

- Relative to standard endogenous-bidder-entry auction papers, the novelty is endogenous **seller** entry with heterogeneity.
- Relative to two-sided-market demand papers, the novelty is using auction-theoretic payoffs to microfound platform externalities.
- Relative to seller-heterogeneity auction papers, the novelty is combining seller heterogeneity with entry and using it for fee-counterfactual welfare analysis.

5 Tables and figures

5.1 Table 1 and Table 2: Fee structure and basic auction facts

TABLE 1
FEE STRUCTURE IN WINE AUCTION DATA

Fee	Bidders/ Sellers	Only If Sold	Notation	Amount/ Rate	For Price Range (£)
Buyer premium	Bidders	Yes	e_B	0	
Seller commission	Sellers	Yes	e_S	.102	≤200
Listing fee	Sellers		e_L	£2.10	200.01–1,500

NOTE.—The platform also charges a reserve price fee that is made up of £0.60 for raising the minimum bid and £0.30 for adding a secret reserve price, but these are not part of the analysis, which focuses on fee structure $e = \{e_B, e_S, e_L\}$. All reported fees include a 20% value-added tax.

TABLE 2
AUCTION-LEVEL DESCRIPTIVE STATISTICS

	<i>N</i>	Mean	Standard Deviation	Minimum	Median	Maximum
Hammer price (£)	3,481	140.33	239.68	1.00	82.24	6,000
Number of bidders	3,481	3.10	2.52	0	3	13
Number of bottles	3,481	3.70	4.23	1	2	72
Is sold	3,481	.64	.48	0	1	1
Price per bottle if sold (£)	2,228	74.81	124.55	.50	35.00	2,200
Sold in bond	3,481	.16	.37	0	0	1
Seller has feedback	3,481	.29	.46	0	0	1
Seller has ratings	3,481	.73	.45	0	1	1
Has any reserve	3,481	.67	.47	0	1	1
Reserve price (£)	2,333	136.62	264.31	1.00	75.00	6,000

NOTE.—The hammer price equals the standing price when the auction closes, irrespective of whether the item is sold. Sold “in bond” indicates that the wine has been stored in a bonded warehouse since arriving in the United Kingdom. Winning bidders can provide textual feedback describing the interaction with the seller and can also rate the interaction as “positive,” “neutral,” or “negative.” Whether the listing has a reserve price includes both secret reserve prices and increased minimum bid amounts.

Read:

- The platform charges a flat listing fee plus a seller commission, but no buyer premium in the observed data.
- The average hammer price is modest relative to flagship auction houses, consistent with BW operating at the lower end of the fine-wine secondary market.
- Two-thirds of listings have a reserve price and about two-thirds sell, so reserve-setting is empirically central rather than a niche feature.

- The bidder count is small on average, which makes entry and thin-market effects quantitatively important.

5.2 Table 3 and Table 4: seller heterogeneity and thin markets

TABLE 3
SELLER-LEVEL DESCRIPTIVE STATISTICS

Statistic	N	Mean	Standard Deviation	Minimum	Median	Maximum
Average number of words in description	254	73.13	83.54	3.00	46.00	772.00
Probability allowing for collection	254	.60	.48	.00	1.00	1.00
Number of sold listings	254	7.43	23.12	.00	2.00	237.00
Number of bottles per listing	254	3.05	5.62	1.00	1.25	72.00
If multiple listings: number of bottles per listing	143	2.42	2.58	1.00	1.43	13.21
If multiple listings: share with $r > 0$	143	.56	.46	.00	.75	1.00
If multiple $r > 0$ listings: share with r secret	94	.54	.48	.00	.75	1.00

NOTE.—Descriptive statistics across sellers based on the full estimation sample. When excluding three frequent sellers, the maximum number of sold listings drops to 92, and the other statistics are nearly identical.

TABLE 4
THIN MARKETS

	Percentile									
	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
Number of times product listed, 4 weeks	1	1	1	1	1	2	2	3	6	37
Number of times product listed, 15 months	1	1	1	1	2	3	4	8	16.2	222
Number of times same title occurs, 15 months	1	1	1	1	1	1	1	1	2	17

NOTE.—This table reports deciles of distributions of the number of times a product or listing title is observed in the sample. In the first row, an observation is a product in each 4-week nonoverlapping interval. Conservative product definitions are used (region $\times$ wine type $\times$ vintage decade), corresponding to high-level filters on the website, and products that do not occur in a month are not counted to avoid the large mass at zero. In the second row, an observation is a product, and in the third row it is the title of the listing; in both cases we are counting how often they occur within the full 15 months of the sample.

Read:

- Sellers are highly heterogeneous and often infrequent users of the platform; about half are observed only once and the median seller sells two items during the sample.
- Sellers vary dramatically in how much information they provide, from very short to extremely long descriptions, consistent with heterogeneity in both observed and unobserved seller traits.
- The market is genuinely thin: many products appear only once in a month, and even over 15 months identical or near-identical listings are rare.
- These facts justify a model with listing inspection costs rather than a commodity-auction model with easy substitution across many close listings.

5.3 Figure 1 and Table 5: model fit and structural estimates

Read:

- The model fits the empirical distributions of second-highest bids, reserve prices, hammer prices, and bidder counts well.
- A particularly demanding fit is that hammer prices in positive-reserve auctions are matched using bidder values estimated from the disjoint sample of zero-reserve auctions.
- The rich set of observables explains a large share of price variation: the R^2 is about 0.55 in the main sample and 0.91 in the high-end sample.
- Estimated inspection costs are economically meaningful: roughly 7% of hammer price per bottle in the main sample and 5% in the high-end sample.
- In positive-reserve auctions, the estimated latent mean bidder count is 4.4 even though only about two actual bidders are observed on average, showing that censoring matters.

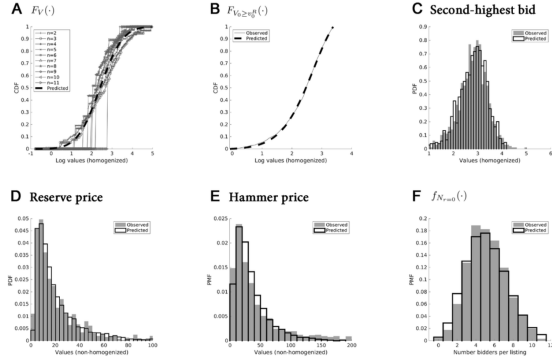


FIG. 1.—Model fit: main sample. Model predictions and observed values of F_i and empirical CDF for $n = 2, \dots, 11$ bidders ($r = 0$ auctions; A), $F_{i,0}$ and empirical CDF ($r > 0$ auctions; B), second-highest bid ($r = 0$ auctions; C), reserve price (prediction includes estimated quality, $r > 0$ auctions; D), hammer price (prediction includes estimated quality, $r > 0$ auctions; E), and number of bidders per listing ($r = 0$ auctions; F). Simulations of bidder values are based on 1,000 draws for each bidder, and simulations of seller values are based on 5,000 draws. Observed values are based on the estimation sample.

TABLE 5
ESTIMATED STRUCTURAL PARAMETERS

	A. Main Sample					
	$\hat{\mu}_b$	$\hat{\sigma}_b$	$\hat{\kappa}_b$	$\hat{\mu}_s$	$\hat{\sigma}_s$	$\hat{\kappa}_s$
Valuation distributions:						
Estimate	2.410	.879	.014	2.574	.757	.211
Standard error	(.025)	(.001)	(.002)	(.026)	(.005)	(.006)
	$\hat{e}_{B,r>0}$	$\hat{e}_{B,r=0}$	$\hat{e}_S$	$\hat{\rho}_{0,r>0}$	$\hat{v}_0^R$	
Entry parameters:						
Estimate	1.810	2.240	2.283	.049	-.080	
Standard error	(.095)	(.121)	(.126)	(.000)	(.039)	
	B. High-End Sample					
	$\hat{\mu}_b$	$\hat{\sigma}_b$	$\hat{\kappa}_b$	$\hat{\mu}_s$	$\hat{\sigma}_s$	$\hat{\kappa}_s$
Valuation distributions:						
Estimate	5.316	.338	-.465	5.495	.359	-.078
~Standard error	(.030)	(.002)	(.008)	(.811)	(1.268)	(.021)
	$\hat{e}_{B,r>0}$	$\hat{e}_{B,r=0}$	$\hat{e}_S$	$\hat{\rho}_{0,r>0}$	$\hat{v}_0^R$	
Entry parameters:						
Estimate	13.782	14.493	14.463	.115	4.705	
~Standard error	(.509)	(.528)	(2.048)	(.004)	(.031)	

NOTE.—The valuation distribution parameters describe the location (μ), scale (σ), and shape (κ) of $\mathcal{G}\mathcal{G}\mathcal{D}$ -distributed log values. For the results in the main sample (panel A), the table reports point estimates and standard errors based on 250 nonparametric bootstrap repetitions. These standard errors are based on draws that generate a stable equilibrium, thereby excluding five draws where the estimator does not converge. Estimation of θ , excludes the 8.2% of sellers for which $\hat{v}_0$ is estimated to be negative. Both $\hat{v}_0$ and $\hat{g}(\mathbf{Z})$ are trimmed at their 1st and 99th percentiles to minimize the impact of outliers; before trimming, the estimation sample has 2,787 observations (618 in the high-end sample). Panel B reports estimation results for the smaller high-end sample, where a larger number of draws (53) does not converge. In this sample, $\hat{v}_0$ is estimated to be negative for 3.7 of sellers, and $\hat{v}_0$ and $\hat{g}(\mathbf{Z})$ are trimmed at their 2nd and 98th percentiles. Excluding a sizable share of bootstraps might bias the standard errors, which is emphasized with the “~Standard error” notation in panel B. See Egedal, Lai, and Su (2015) and Aguirregabiria and Marcoux (2021) on the issue of nonconvergence of the NPL estimator in small samples based on stable data-generating processes.

5.4 Table 6: estimated indirect network effects

Read:

- Bidders generate stronger positive externalities for sellers than sellers generate for bidders.
- Adding 10 bidders raises the expected surplus of the marginal seller by about £0.007 in the main sample, more than twice the effect of adding 10 sellers on bidder surplus.
- Seller selection materially dampens the positive effect of extra sellers on bidders. Adding 50 sellers raises bidder surplus much less when the newly entering sellers are the marginal sellers implied by the model than when sellers are homogeneous.
- The high-end sample shows even larger nonlinearities, reinforcing the importance of heterogeneity.

TABLE 6
ESTIMATED INDIRECT NETWORK EFFECTS

	Main Sample				High-End	
	Change of -50	Change of -10	Change of +10	Change of +50	Change of -50	Change of +50
A. Sellers ($T_{s,0}$)						
Effect on $\Pi_{B,r>0}$ (selection)	-.012	-.003	.003	.017	-.446	.597
Effect on $\Pi_{B,r=0}$ (no selection)	-.034	-.007	.007	.033	-.741	.688
B. Bidders ($M_{b,0}$)						
Effect on $\Pi_{S,r>0}$ (marginal seller)	-.033	-.007	.007	.033	-.437	.437
Effect on $\Pi_{S,r=0}$ (median seller)	-.057	-.011	.011	.057	-1.343	1.335

NOTE.—Simulations are based on $r > 0$ homogenized auctions. In the main (high-end) sample, $T_{s,0}$ equals 1,589 (434), $M_{b,0}$ equals 8,131 (2,279), and the marginal seller has a valuation at the 83rd (96th) percentile of the estimated seller valuation distribution.

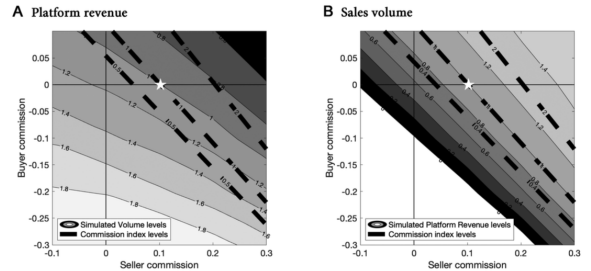


FIG. 2.—Illustration of commission index and revenue-volume trade-off. Two overlapping contour plots are displayed. The gray contour plot reflects simulated platform revenues normalized by baseline revenues (A) or the simulated volume of sales normalized by the baseline volume of sales (B). The thick dashed lines indicate combinations of c_b and c_s where the commission index is either about half as large as (level = 0.5 for revenue, level = 0.4 for volume), equal to (level = 1), or twice as large as (level = 2) the baseline commission index. The game is estimated on a grid of $c_b \times c_s$ ($c_b = \{-0.3, -0.2, -0.1, 0, 0.1\}$, $c_s = \{-0.1, 0, 0.1, 0.2, 0.3\}$) and is interpolated linearly. The white star indicates the baseline fees. Simulations are based on parameter estimates from the main sample.

5.5 Figure 2: commission index and the revenue-volume trade-off

Read:

- The relevant fee object for platform revenue is the commission index

$$a = \frac{c_B + c_S}{1 + c_B}.$$

- Holding the commission index fixed leaves platform revenue unchanged, but not sales volume.
- Moving fee burden from sellers to buyers can shrink volume even if revenue is unchanged, because bidders shade bids when the buyer premium rises.
- The platform therefore faces a genuine revenue-volume trade-off, not just a one-dimensional pricing problem.

5.6 Figure 3: lemons effect and heterogeneous seller incidence

Read:

- When the listing fee increases by £1, sellers who remain on the platform lose much less than £1 because some high-value sellers exit.
- The effect is heterogeneous across sellers: the surplus reduction is 22–46 percentage points smaller than in a model with no entry response.
- The bias grows when seller heterogeneity is made larger, which is direct evidence that selection is the economic mechanism.
- Using the proceeds to subsidize bidder entry can raise the surplus of inframarginal sellers by up to about £1.20 despite the higher listing fee.

5.7 Figure 4: negative buyer commission and platform profit maximization

Read:

- A no-subsidy restriction on buyer fees is binding in the revenue-maximization problem.
- Platform revenues can rise only if buyers are subsidized through a negative buyer commission, financed by higher seller-side fees.
- Along the volume constraint, the paper finds fee reallocations that can increase revenues by up to about 40%.
- This is a clean two-sided-market lesson: below-marginal-cost pricing on one side can be profit-maximizing because that side creates stronger indirect network effects.

5.8 Antitrust counterfactuals

Read:

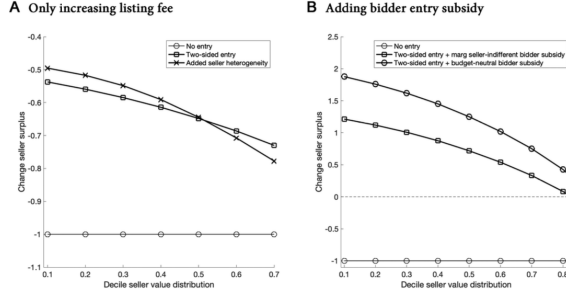


FIG. 3.—Lemons effect: heterogeneous change in seller surplus when increasing the listing fee by £1. The estimated effects are plotted by decile of $F_{i|W>0}^s$ for sellers who are inframarginal (with $v_b \in [v_b^l, v_b^h]$) both at baseline and in the counterfactual. Simulations are based on $r > 0$ homogenized auctions in the main sample.

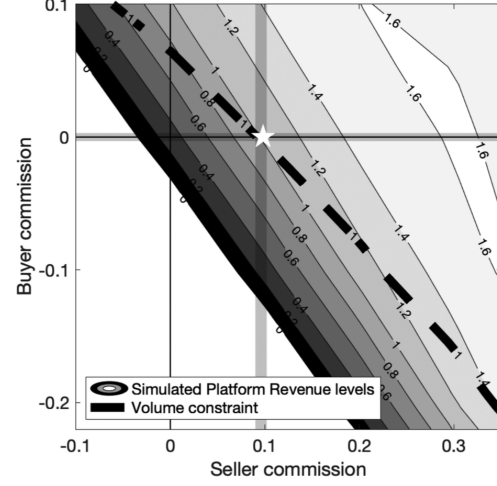


FIG. 4.—Platform revenue at alternative fee structures. Shown is a contour plot of simulated platform revenues normalized by baseline values; also indicated is the volume constraint where the simulated volume of sales is equal to the baseline volume (thick dashed line). The gray vertical bar corresponds to $c_s \in [0.9 \text{ (high-end)}, 0.102 \text{ (main sample)}]$; the horizontal bar indicates the baseline $c_b = 0$. The game is estimated on a grid of $c_b \times c_s$ ($c_b = \{-0.3, -0.2, -0.1, 0, 0.1\}$, $c_s = \{-0.1, 0, 0.1, 0.2, 0.3\}$), and values are interpolated linearly. The white star indicates the baseline fees. Results are based on parameter estimates from both main and high-end samples.

- When the commission index is doubled from 0.102 to 0.204, only about 60% of the seller-commission incidence falls on sellers, far below the 100% predicted by one-sided benchmark rules.
- Accounting for two-sided entry raises total welfare losses for sellers by about 6.5% relative to simpler models.
- Buyers are harmed as well, including winning bidders, contrary to some conventional assumptions in antitrust practice.
- The paper uses the Sotheby's-Christie's commission-fixing case to show why fee-damage calculations need structural two-sided models.

6 Short summary

- **Conclusion.** The paper estimates a structural auction platform model in which bidder and seller entry are jointly determined.
- **Identification insight.** Bidder values come from zero-reserve auctions, seller values come from reserve-price inversion, and entry costs come from equilibrium zero-profit conditions.
- **Economic lesson.** In two-sided auction platforms, seller selection changes the composition of listings, bidder entry, prices, and welfare. This makes fee incidence nonlinear and often very different from one-sided intuition.
- **Policy lesson.** Platform pricing and antitrust damages cannot be evaluated using direct overcharges alone. They must account for equilibrium responses on both sides of the market.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that a two-sided auction platform with endogenous bidder and seller entry can be estimated structurally with relatively standard auction data.
- It also establishes that seller heterogeneity and selective seller entry are not second-order complications: they fundamentally alter fee incidence, revenue, and welfare analysis.
- The empirical application shows that the model fits well and that the implied counterfactuals are economically important.

7.2 Economic intuition

In a one-sided auction environment, higher seller fees simply tax sellers. Here they do more: they change *which* sellers choose to list. Since higher-value sellers set higher reserve prices, excluding some of them makes the remaining listings more attractive to bidders. This encourages bidder entry, increases competition in the remaining auctions, and partly offsets the seller-side harm from the fee increase. That is the paper’s lemons effect. Once that channel is present, fee incidence and optimal pricing become equilibrium objects shaped by both sides of the market.

7.3 Takeaway

This paper is best read as a structural bridge between auction theory and two-sided platform economics. Its main message is that platform fees cannot be interpreted mechanically. In markets with endogenous participation on both sides, the platform changes not only the amount of activity but also the *composition* of participants. That composition effect can overturn standard intuitions about who pays, who gains, and what constitutes harm. For any empirical project on online auctions, marketplace design, peer-to-peer platforms, or two-sided antitrust, this paper is a benchmark reference.